

Assignment 1; due Thursday, September 24, 2026

- [1] Let $E \subseteq \mathbb{R}$ be nonempty. We say that $t \in \mathbb{R}$ is a *lower bound* of E if $t \leq x$ for all $x \in E$. We say that E is *bounded below* if there exists a lower bound of E . We say that $t \in \mathbb{R}$ is a *greatest lower bound*, or *infimum*, of E if it is a lower bound of E and if $t \geq s$ for any lower bound s of E . If E has an infimum, it is denoted $\inf E$.
- [a] Let E be a nonempty subset of $\mathbb{R}$ which is bounded below. Find a way to use the completeness axiom to prove that $\inf E$ exists. *Hint*: Consider the subset $-E = \{-x : x \in E\}$.
- [b] Let $\emptyset \neq A \subseteq B \subseteq \mathbb{R}$. Suppose B is bounded below (so $\inf B$ exists). Prove that A is bounded below (so $\inf A$ exists) and that $\inf A \geq \inf B$.
- [c] Let A, B be nonempty subsets of $\mathbb{R}$, both bounded below (so both $\inf A$ and $\inf B$ exist.) Define

$$A + B = \{a + b : a \in A, b \in B\}.$$

Prove that $A + B$ is bounded below and that $\inf(A + B) = \inf A + \inf B$.

- [2] Let x, y be two real numbers such that $y - x > 1$. (In particular, $x < y$.)
- [a] If x is positive, prove that there exists an integer $k \in \mathbb{Z}$ such that $x < k < y$.
- [b] If x is negative and y is positive, prove that there exists an integer $k \in \mathbb{Z}$ such that $x < k < y$.
- [c] If y is negative, prove that there exists an integer $k \in \mathbb{Z}$ such that $x < k < y$.

- [3] The parts of this problem are *not* related.
- [a] Let $x \in \mathbb{R}$. Consider the set $E = \{q \in \mathbb{Q} : q < x\} \subset \mathbb{Q} \subset \mathbb{R}$. Show that E is nonempty and bounded above, so $\sup E$ exists. Show also that $\sup E = x$.
- [b] Let $x, y \in \mathbb{R}$ with $x < y$. Prove that there exists an *irrational* (not rational) number z such that $x < z < y$. *Hint*: You may use the fact that $\sqrt{2}$ is irrational.

- [4] Let $C > 0$ be a fixed positive number.
- [a] Let $x \geq 0$, and suppose that for any $\varepsilon > 0$, we have $x < C\varepsilon$. Show that for any $\eta > 0$, we have $x < \eta$.
- [b] In the situation of part [a], show that we must have $x = 0$. *Hint*: Find a way to use the Archimedean property.

- [5] In this problem you will give an *alternative proof* of the Cauchy-Schwarz inequality. The inequality says that

$$-\|\mathbf{x}\|\|\mathbf{y}\| \leq \langle \mathbf{x}, \mathbf{y} \rangle \leq \|\mathbf{x}\|\|\mathbf{y}\|.$$

Both inequalities are trivially seen to be equalities if either $\mathbf{x} = \mathbf{0}$ or $\mathbf{y} = \mathbf{0}$, so we suppose that both vectors are nonzero.

By consider the norm of the vector $\mathbf{z} = \frac{\mathbf{x}}{\|\mathbf{x}\|} \pm \frac{\mathbf{y}}{\|\mathbf{y}\|}$, show that

- [a] both inequalities are true,
- [b] the first inequality becomes an equality if and only if $\mathbf{y} = t\mathbf{x}$ for some $t < 0$,
- [c] the second inequality becomes an equality if and only if $\mathbf{y} = t\mathbf{x}$ for some $t > 0$.

- [6] Let $E = \overline{B_r(\mathbf{x}_0)} = \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x} - \mathbf{x}_0\| \leq r\}$ be a closed ball. Prove that E is *not open*.

[7] Let $F = \{\mathbf{x}_1, \dots, \mathbf{x}_N\}$ be a finite subset of $\mathbb{R}^n$. Prove that F is *closed*.

[8] Let $\mathbf{a} \in \mathbb{R}^n \setminus \{\mathbf{0}\}$, and $c \in \mathbb{R}$. Consider the subset $E = \{\mathbf{x} \in \mathbb{R}^n : \langle \mathbf{x}, \mathbf{a} \rangle < c\}$. Prove that E is open.
Hint: Observe that $|\langle \mathbf{y}, \mathbf{a} \rangle - \langle \mathbf{z}, \mathbf{a} \rangle| = |\langle \mathbf{y} - \mathbf{z}, \mathbf{a} \rangle| \leq \|\mathbf{y} - \mathbf{z}\| \|\mathbf{a}\|$ by Cauchy–Schwarz.

[9] Let $C \subset \mathbb{R}^n$ be a non-empty proper *closed* subset of $\mathbb{R}^n$. Let $\mathbf{x} \in \mathbb{R}^n \setminus C$, which exists because C is a proper subset. Define the subset $E \subset \mathbb{R}$ by

$$E = \{r > 0 : B_r(\mathbf{x}) \cap C = \emptyset\}.$$

[a] Explain why $E \neq \emptyset$

[b] Prove that there exists an upper bound for E .

[c] Thus, by the completeness axiom, $\sup E$ exists. Show that $\sup E \leq \|\mathbf{x} - \mathbf{y}\|$ for all $\mathbf{y} \in C$.

[10] Let $(\mathbf{x}_k)_{k \in \mathbb{N}}$ be a sequence in $\mathbb{R}^n$, and suppose that there exists $r \in (0, 1)$ and $C > 0$ such that

$$\|\mathbf{x}_{k+1} - \mathbf{x}_k\| \leq Cr^k.$$

Prove that $(\mathbf{x}_k)$ is Cauchy.

Hint: Recall that $r^{k+m-1} + r^{k+m-2} + \dots + r^k = r^k(r^{m-1} + r^{m-2} + \dots + 1) = r^k \frac{1-r^m}{1-r}$.

You may also use without proof the fact (which is easy to show using the Archimedean Property) that: since $r \in (0, 1)$, given any $\delta > 0$, there exists $N \in \mathbb{N}$ such that $k \geq N$ implies $r^k < \delta$.
